



Assam Science and Technology University

Department of Computer Science & Engineering
Guwahati, Assam, India

AI-Based Crop Recommendation System for Assam using Real-Time Weather Data

Submitted by:

Name	Enrollment Number	Branch
Anoushka Goswami	2481106018	Data Science
Jessica Sonowal	2481106019	Data Science
Nayan Nirban Dewri	2481108007	Data Science
Nibir Niloy Kalita	2481108008	Data Science

Under the Guidance of:

Miss Sanjukta Sarma, Assistant Professor
Department of Computer Science & Engineering

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Submitted to the Assam Science and Technology University (ASTU)

Declaration

i. Statement by the Students

We, the undersigned students of Assam Science and Technology University, hereby solemnly declare that the Micro Project Report entitled “**AI-Based Crop Recommendation System for Assam using Real-Time Weather Data**” submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering, is a bonafide record of original work carried out by us under the guidance of **Miss Sanjukta Sarma**, Assistant Professor, Department of Computer Science & Engineering.

We further declare that:

1. The project and the work presented in this report are entirely our own and have not been submitted, in whole or in part, for the award of any other degree or diploma at this institution or any other institution.
2. All sources of information used in this report have been duly acknowledged through proper citation and references.
3. We have not copied or reproduced any portion of this work from any published or unpublished source without due acknowledgement.
4. The system developed, including the machine learning model, data mappings, API integration, and user interface, represents original work developed independently by our team.
5. All experimental data, model accuracy metrics, and market figures reported herein are accurate to the best of our knowledge.

Place: Guwahati, Assam

Date: _____

Anoushka Goswami
Enrollment No.: 2481106018

Jessica Sonowal
Enrollment No.: 2481106019

Nayan Nirban Dewri
Enrollment No.: 2481108007

Nibir Niloy Kalita
Enrollment No.: 2481108008

Certified by:

Miss Sanjukta Sarma
Assistant Professor
Dept. of CSE
ASTU

Dr. Mridusmita Barooah
Head of Department
Dept. of CSE
ASTU

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Miss Sanjukta Sarma

Assistant Professor, Dept. of CSE

ASTU

Acknowledgement

i. Acknowledgement to Dean and Guide

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Anoushka Goswami
Roll No.: 2481106018

Jessica Sonowal
Roll No.: 2481106019

Nayan Nirban Dewri
Roll No.: 2481108007

Nibir Niloy Kalita
Roll No.: 2481108008

ii. Acknowledgement to the Institution

We are profoundly grateful to **Assam Science and Technology University**, Guwahati, Assam, for providing us with a world-class academic and infrastructural environment. The institution's commitment to practical, project-based learning gave us the opportunity to develop a real-world application that addresses genuine challenges faced by the farming community of Assam.

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Anoushka Goswami
Roll No.: 2481106018

Jessica Sonowal
Roll No.: 2481106019

Nayan Nirban Dewri
Roll No.: 2481108007

Nibir Niloy Kalita
Roll No.: 2481108008

Abstract

Agriculture remains the principal livelihood of over 53% of Assam’s workforce, yet a persistent gap exists between modern agronomic science and actual farm-level decision-making. This micro project presents the design, development, and deployment of an **AI-Based Crop Recommendation System** specifically engineered for Assam’s agro-ecological conditions, integrating supervised machine learning with real-time meteorological data retrieval.

The core recommendation engine is a **Random Forest Classifier** trained on a dataset of 5,000+ records comprising seven agronomic features: Nitrogen (N), Phosphorus (P), Potassium (K), soil pH, ambient temperature, relative humidity, and monthly rainfall. The model achieved a classification accuracy of **97.3%** on a held-out 20% test set, outperforming Decision Tree (91.4%) and XGBoost (96.8%) baselines across all evaluation metrics — precision, recall, and F1 score — on a **49-class** crop label space that includes indigenous Assam varieties such as Joha Rice, Komal Saul, Bora Rice, Bhut Jolokia, and Assam Tea.

Live weather parameters (temperature: T , humidity: H , rainfall: R) are automatically retrieved from the **OpenWeatherMap REST API** for the farmer’s selected district, eliminating the need for manual environmental data entry. Soil macronutrient values are abstracted from a simple soil-type selector using ICAR-sourced regional baseline tables, reducing the technical input burden from seven continuous parameters to three categorical choices.

Beyond basic recommendation, the system incorporates three advanced analytical modules: (i) a **Fertilizer Prescription Calculator** using ICAR-standard NPK deficit formulas to output exact Urea, DAP, and MOP application rates in kg/hectare; (ii) a **Market Economics Forecaster** that computes per-hectare gross revenue, input cost, net profit, and return on investment (ROI) from Assam Mandi price data; and (iii) a **Cloud-Based Vision AI** camera interface powered by the **Google Gemini API**, which applies large multimodal model reasoning for high-precision botanical identification of crop species and rare indigenous varieties — including Bhut Jolokia and Bao Rice — directly from farmer-uploaded photographs. Additionally, all recommendations are securely logged in a persistent **SQLite Database** for historical tracking and regional analytics.

Accessibility is addressed through a bilingual interface supporting English and *Assamese* (), a browser-native **Text-to-Speech** player, and automated **PDF report generation** with Unicode font support. The application is deployed as a responsive web interface built with **Streamlit** on Streamlit Community Cloud, achieving an average end-to-end response time of under two seconds.

Keywords: Machine Learning, Random Forest, Crop Recommendation, Assam Agriculture, OpenWeatherMap API, Gemini Vision AI, Fertilizer Prescription, Market Forecasting, Streamlit, NPK Estimation, Bilingual Interface, Regional Analytics.

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Chapter 1: Introduction

1.1 Background of the Project

Agriculture is the cornerstone of Assam's socio-economic fabric. The state, situated in the north-eastern region of India, is endowed with fertile alluvial plains formed by the Brahmaputra river system, a humid subtropical climate, and abundant annual rainfall averaging 1,800–2,000 mm. These conditions collectively support one of the most diverse agricultural landscapes in the country, sustaining major crops such as rice, jute, tea, mustard, and sugarcane alongside a rich repertoire of indigenous varieties including Joha Rice (GI-tagged aromatic rice), Komal Saul (soft pre-cooked rice), Bhut Jolokia (the world's once-hottest chilli), and Assam Tea.

Despite this agricultural richness, the farming community of Assam faces persistent structural challenges. According to the Agricultural Statistics of Assam 2023–24 published by the Directorate of Agriculture, Government of Assam, average crop yields in the state remain 15–25% below national averages across major categories. A primary contributing factor is the absence of scientific, data-driven crop selection at the farm level. Most farmers — particularly in rural districts — rely on generational experience, peer advice, or seasonal tradition when deciding which crop to cultivate. This empirical approach, while rich in contextual knowledge, lacks the capacity to adapt to year-on-year variations in soil health, weather patterns, and market dynamics.

Simultaneously, the digital revolution in Indian agriculture has produced a range of mobile and web-based advisory platforms. However, most existing systems suffer from critical limitations when applied to Assam's specific context: they require users to manually input precise soil nutrient values (NPK concentrations, pH) obtainable only through laboratory soil tests that cost Rs. 500–1,500 and take 7–14 days to process — a barrier that effectively excludes the majority of small and marginal farmers who constitute over 80% of Assam's farming population.

The convergence of affordable machine learning tools, freely available meteorological APIs, accessible web deployment platforms, and powerful cloud-based multimodal AI creates a unique opportunity to bridge this gap. This project is conceived and executed within that opportunity space.

1.2 Problem Statement

Despite Assam's significant agricultural potential, farmers in the state lack access to an intelligent, accessible, and regionally specific decision-support tool that:

- (i) recommends the most suitable crop for a given set of environmental and soil conditions without requiring laboratory-measured input parameters;
- (ii) accounts for the full diversity of Assam-specific crop varieties, including indigenous varieties absent from national databases;
- (iii) dynamically incorporates real-time weather data to ensure recommendations are contextually accurate on the day of use;
- (iv) provides actionable downstream guidance on fertilizer application and expected market profitability; and
- (v) is accessible to users with limited English literacy through a regional language interface.

The consequence of this gap is measurable: suboptimal crop selection contributes to yield losses, soil degradation from inappropriate cultivation, underutilisation of high-value indigenous varieties, and avoidable financial loss for farming households. This project proposes and implements a system that directly addresses each of the above dimensions.

1.3 Objectives of the Project

The primary objectives of this project are as follows:

- 1. Develop a high-accuracy ML recommendation engine:** Design and train a supervised machine learning classification model capable of recommending optimal crops from soil and environmental parameters with a target accuracy of $\geq 95\%$ on Assam-specific crop data.
- 2. Cover 49 Assam-specific crop varieties:** Extend the standard crop dataset to include indigenous varieties specific to Assam's agro-ecological zones, including eight distinct rice varieties (Joha, Bao, Bora, Chokuwa, Komal Saul, Xaaj, Apong, Joha-72).
- 3. Integrate real-time weather data:** Automatically retrieve live temperature, humidity, and rainfall readings from the OpenWeatherMap API for any of Assam's 35 districts.
- 4. Minimise technical input requirements:** Abstract NPK and pH values from a simple soil-type selector using ICAR-validated regional baseline data, eliminating the dependency on laboratory soil testing.
- 5. Implement a fertilizer prescription engine:** Calculate exact fertilizer application rates (Urea, DAP, MOP) using ICAR-standard NPK deficit formulas to correct soil nutrient deficiencies for the recommended crop.
- 6. Provide market economics forecasting:** Display per-hectare gross revenue, estimated input cost, net profit, and ROI based on current Assam Mandi market prices.

7. **Enable cloud-based Vision AI for crop identification:** Integrate the Google Gemini API to perform multimodal botanical reasoning on farmer-uploaded photographs, enabling high-precision identification of both common and rare indigenous Assam crop species without requiring on-device model weights.
8. **Ensure bilingual accessibility:** Implement a full English–Assamese () language toggle with text-to-speech output for low-literacy users.
9. **Deploy as a production web application:** Host the complete system on Streamlit Community Cloud, accessible via a public URL with no installation required.

1.4 Scope of the Project

1.4.1 Geographical Scope

The system is designed exclusively for the state of Assam, India. Weather data retrieval covers all 35 administrative districts of Assam, mapped to their respective nearest city names recognised by the OpenWeatherMap API. Soil type classifications and NPK baseline values are derived from ICAR data specific to Assam’s major agro-climatic zones: the Brahmaputra Valley Zone, the Barak Valley Zone, and the Hill Zone.

1.4.2 Crop Scope

The recommendation model covers **49 crop varieties**, categorised as:

- **Standard field crops:** Rice, wheat, maize, jute, mustard, sugarcane, cotton (7 crops)
- **Pulse and legume crops:** Chickpea, lentil, mung bean, moth beans, pigeon peas, kidney beans, black gram (7 crops)
- **Horticultural crops:** Banana, mango, papaya, coconut, pomegranate, grapes, watermelon, muskmelon, orange, apple, coffee (11 crops)
- **Assam indigenous rice varieties:** Joha Rice, Joha-72, Bao Rice, Bora Rice, Chokuwa Rice, Komal Saul, Xaaj Rice, Apong Rice (8 varieties)
- **Assam-specific fruits and vegetables:** Bhut Jolokia, Kaji Nemu, Ou Tenga, Thekera, Amlokhi, Bogori, Leteku, Jalpai, Areca Nut, Betel Leaf, Kola Banana (11 crops)
- **Assam leafy greens and specialty crops:** Lai Xaak, Kosu, Dhekia Xaak, Manimuni, Local Brinjal, Local Pumpkin, Local Beans, Bamboo Shoot, Assam Tea (9 crops)

1.4.3 Functional Scope

The system delivers the following functional outputs:

- Ranked list of top n recommended crops with AI confidence percentages

- Real-time weather display for the selected district
- Fertilizer prescription (Urea, DAP, MOP in kg/hectare)
- Market economics forecast (revenue, cost, net profit, ROI)
- Cloud-based Vision AI crop identification via Google Gemini API
- Text-to-speech recommendation summary
- Downloadable PDF report in English or Assamese

1.4.4 Technical Scope

The system is built as a highly modular web application using Python and Streamlit. It requires an internet connection for weather API calls, Google Gemini Vision AI inference, and Wikipedia image retrieval. The machine learning recommendation model runs server-side. The Vision AI module leverages the Google Gemini multimodal API, enabling powerful cloud-based botanical reasoning that scales to rare indigenous varieties without bundling heavyweight model weights in the deployment package. All generated reports are stored persistently using a local SQLite database architecture. Voice output relies on the browser's built-in Web Speech API.

1.4.5 Limitations and Out-of-Scope Items

The following are explicitly outside the current scope:

- Irrigation scheduling or water management recommendations
- Real-time Mandi price feeds (current version uses static averages)
- Offline/mobile-native Android or iOS application
- IoT soil sensor hardware integration
- Multi-state (beyond Assam) coverage
- Hindi or Bengali language support

Chapter 2: Literature Review

2.1 Summary of Existing Work Related to the Project

The application of machine learning to precision agriculture has been an active area of research over the past two decades, producing a wide range of crop recommendation and yield prediction systems. This chapter provides a comprehensive review of the most significant published works directly relevant to the present project, covering algorithmic approaches, hardware integration, data collection strategies, and system deployment methodologies.

2.1.1 Ensemble-Based Crop Recommendation Using Majority Voting

One of the earliest and most frequently cited works in the domain of machine-learning-driven crop recommendation is that of Reddy, Dadore, and Watekar (2019), who developed a system specifically for the Ramtek region of Maharashtra. Their method collected soil parameters—depth, texture, pH, soil colour, permeability, drainage, water holding capacity, and erosion—from the local soil testing laboratory in Nagpur. The study employed an ensemble model using majority voting that combined four base learners: Random Tree, CHAID (Chi-squared Automatic Interaction Detection), K-Nearest Neighbour (KNN), and Naive Bayes. The system was implemented in RapidMiner and generated IF-THEN production rules for crop prediction. A representative rule from their system reads: if pH is mid-alkaline, depth is above 100, water holding capacity is low, drainage is moderately well, and erosion is moderate, then the recommended crop is Paddy. Their work demonstrated the viability of ensemble models for agricultural data and highlighted that combining multiple weaker learners through voting consistently outperforms any single classifier when agricultural features exhibit complex, non-linear interactions.

Building directly on this approach, Rajak et al. (2017) from Sinhgad Academy of Engineering developed a parallel ensemble system that substituted Support Vector Machine (SVM) and Artificial Neural Network (ANN) as base learners within the same majority-voting framework. Their dataset was collected from Polytest Laboratories soil testing lab in Pune and covered nine crop categories including groundnut, pulses, cotton, banana, paddy, sorghum, and sugarcane. The study provided detailed mathematical formulations of each learner including the SVM hyperplane equation and the ANN multi-layer perceptron propagation model, and validated the ensemble against individual classifiers. Their finding that the ensemble consistently achieved higher accuracy than any single component learner corroborated the theoretical advantage of model combination through diversity and complementarity.

2.1.2 Multi-Algorithm Comparative Studies on Standard Datasets

A landmark comparative study was published by Gosai et al. (2021) from Charotar University of Science and Technology, which evaluated six machine learning algorithms—Decision Tree, Naive Bayes, SVM, Logistic Regression, Random Forest, and XGBoost—on the Kaggle Crop Recommendation Dataset containing 2,200 instances across 22 crop classes. The feature set comprised Nitrogen (N), Phosphorus (P), Potassium (K), soil pH, humidity, temperature, and rainfall—the same feature vector adopted in the present project. The study's experimental results showed that XGBoost achieved the highest accuracy at 99.31%, followed closely by Naive Bayes and Random Forest at 99%, while SVM performed significantly below expectation at only 10.68% accuracy due to its kernel sensitivity to the particular data distribution. The authors concluded that tree-based ensemble methods generalise most reliably to this type of multi-class agricultural classification problem. Importantly, the study also proposed a five-stage pipeline—data collection, preprocessing, feature extraction, algorithm application, and recommendation output—that closely mirrors the architecture implemented in the present project.

The study by Apat, Mishra, Raju, and Padhy (2023), published in the *Journal of Scientific and Industrial Research*, introduced a two-phase Classification-Regression-Boosting (CRB) model applied to crop recommendation. Phase one applied the Synthetic Minority Over-sampling Technique (SMOTE) to balance the inherently skewed crop dataset before training, and phase two evaluated 14 distinct classifiers spanning the classification, regression, and boosting families. Their results showed that CatBoost (CBoost) achieved the highest overall accuracy of 99.51%, F-measure of 0.9916, and precision of 0.9918, while Gaussian Naive Bayes outperformed all others specifically on the ROC (0.9569) and MCC (0.9569) metrics. Their key finding that data balancing through SMOTE significantly improves recall for minority crop classes is directly applicable to Assam-specific crops such as Joha Rice and Bhut Jolokia, which are substantially underrepresented in national datasets.

2.1.3 AI and IoT Integration for Real-Time Soil Data Collection

Bandara, Weerasooriya, Ruchirawya, Nanayakkara, Dimantha, and Pabasara (2020) from the Sri Lanka Institute of Information Technology proposed a holistic crop recommendation architecture that integrated physical IoT hardware with machine learning software. Their system employed four Arduino-connected sensors—sunlight intensity, soil moisture, soil pH, and a combined temperature-humidity sensor—to automatically collect real-time environmental data without manual input. The overall system accuracy was reported at 92%, with specific module accuracies of 95% for the soil and sunlight detection model, and 96% for the feedback system. Their study also implemented natural language processing (NLP) for Sinhala-language farmer feedback, directly demonstrating the feasibility of regional language integration in agricultural AI—a concept extended in the present project through the Assamese language toggle.

Pathak, Gurjar, Jaiswal, and Shaikh (2025) from AITR Indore independently identified a similar architectural pattern for smart crop recommendation: a user input module, an IoT sensing layer for automated data collection, an AI processing unit based on Random Forest, and a web interface for output delivery. Their user testing confirmed that farmers with limited technical backgrounds could effectively interact with the interface—a critical design validation for accessibility-focused deployment.

2.1.4 Assam-Specific Machine Learning Crop Recommendation

The work most directly aligned with the geographic context of the present project is that of Bhuyan, Patgiri, Medhi, Patel, and Abonmai (2023) from Assam Agricultural University, who developed a machine-learning-based crop recommendation system specifically for the Biswanath district of Assam. Their study collected 180 soil samples across five cropping systems and analysed 13 physical soil properties. Using coefficient of variation (CV) analysis, they identified silt content (CV = 38.87%), clay content (CV = 42.42%), and hydraulic conductivity (CV = 42.49%) as the most spatially variable soil indicators in the region. A Decision Tree Classifier was trained on the physical property dataset and achieved an accuracy of 94%. This study is of exceptional relevance as it provides ground-truthed soil data for Assam's North Bank Plain Zone and validates the use of tree-based classifiers for the region's specific soil variability.

2.1.5 Web-Deployed Crop Recommendation with Flask and Streamlit

Aarthi, Manimegalai, and Sakthivel (2025) from Tamil Nadu Agricultural University published one of the most recent and methodologically rigorous studies in this domain, proposing an AI-based smart crop recommendation system deployed as a Flask web application. Their study evaluated seven algorithms and found Random Forest to be the top performer at 98.2% accuracy, followed by XGBoost at 97.5%, KNN at 94.3%, SVM at 93.4%, Decision Tree at 91.5%, Logistic Regression at 85.6%, and Naive Bayes at 80.1%. Their recommendation for future integration with live weather APIs and multilingual support was directly addressed in the present project.

Akkem, Kumar, and Varanasi (2023) from the National Institute of Technology Silchar published a comprehensive review and implementation study on Streamlit-based applications for advanced ensemble learning in crop recommendation. Their work evaluated three advanced ensemble architectures: stacking (accuracy: 95.5%), ensemble of ensembles (97.8%), and federated ensemble learning (98.9%). The paper is significant for its direct use of the Streamlit framework—the same platform used in the present project—and its demonstration that Streamlit enables a 20% increase in user interaction efficiency over traditional web frameworks for agricultural applications.

2.2 Comparison of Different Approaches

Based on the reviewed papers, Table 2.1 presents a structured comparison of all key systems across the dimensions most relevant to the present project.

Table 2.1: Comprehensive Comparison of Reviewed Crop Recommendation Systems

Study	Algorithm(s)	Best Accuracy	Crops Covered	Live Weather	Regional Language	Deployment Platform
Reddy et al. (2019)	Ensemble: RT, CHAID, KNN, NB	Not reported	7	No	No	RapidMiner
Rajak et al. (2017)	Ensemble: SVM, ANN, RF, NB	Not reported	9	No	No	RapidMiner
Gosai et al. (2021)	DT, NB, SVM, LR, RF, XGBoost	99.31%	22	No	No	Not deployed
Apat et al. (2023)	RF, XGBoost, CBoost, SVM	99.51%	22	No	No	Not deployed
Bandara et al. (2020)	NB, SVM + IoT	92%	4	No (sensor)	Yes (Sinhala)	Mobile App
Pathak et al. (2025)	RF, Rule-based	85–90%	Regional	Proposed	No	Flask
Bhuyan et al. (2023)	Decision Tree	94%	5 (Assam)	No	No	Flask (local)
Aarathi et al. (2025)	DT, KNN, SVM, RF, XGBoost	98.2%	22	No	No	Render.com
Akkem et al. (2023)	Stacking, Federated Ensemble	98.9%	22+	No	No	Streamlit
Present Project	Random Forest	97.3%	49 (Assam)	Yes (OWM)	Yes (Assamese)	Streamlit Cloud

The comparison reveals a clear progression in the sophistication of crop recommendation systems over the reviewed period. The present project occupies a distinct position: it is the only reviewed system that simultaneously achieves high accuracy, covers Assam's indigenous crop diversity, integrates live meteorological data, provides a regional language interface, and is deployed as a public-access web application.

2.3 Gaps in Existing Systems

2.3.1 Gap 1: Absence of Assam-Specific and Indigenous Crop Coverage

All reviewed systems except Bhuyan et al. (2023) were trained on datasets covering 4 to 22 crops, none of which included the GI-tagged indigenous varieties that are economically and culturally central to Assam's agricultural identity. Varieties such as Joha Rice (Rs. 6,000/quintal vs. Rs. 2,200 for common rice), Komal Saul, Bora Rice, and Bhut Jolokia (Rs. 25,000/quintal) are entirely absent from all standard datasets and from all reviewed systems. The present project addresses this gap through a custom extended dataset covering 49 Assam-specific crops, including eight distinct indigenous rice varieties and eleven Assam-endemic fruits and vegetables.

2.3.2 Gap 2: Reliance on Laboratory-Measured Soil Parameters

Every reviewed system requires users to manually enter exact values for Nitrogen, Phosphorus, Potassium, and pH—parameters obtainable only after laboratory soil testing costing Rs. 500–1,500 per test and taking 7–14 days. The present project addresses this gap through a Simple Mode in which farmers select their soil type from a dropdown and the system maps this choice to ICAR-validated regional baseline NPK values, eliminating the laboratory dependency entirely.

2.3.3 Gap 3: No Integration of Real-Time Meteorological Data

Despite frequent acknowledgement that temperature, humidity, and rainfall are among the most impactful variables for crop selection, none of the reviewed systems integrates live weather data. Aarthi et al. (2025) explicitly identify weather API integration as a future enhancement. The present project resolves this gap by automatically retrieving temperature and humidity from the OpenWeatherMap REST API for any of Assam's 35 districts.

2.3.4 Gap 4: Absence of Prescriptive Post-Recommendation Analytics

All reviewed systems terminate at the crop recommendation step, providing the farmer with a crop name but no guidance on what to do next. None of the reviewed papers includes a fertilizer prescription module, a market price forecast, or a profitability analysis. The present project directly addresses this gap with two prescriptive modules: a fertilizer prescription calculator using ICAR-standard NPK deficit formulas, and a market economics forecaster computing revenue, cost, net profit, and ROI from Assam Mandi price data.

2.3.5 Gap 5: No Regional Language or Accessibility Support

With the exception of Bandara et al. (2020), who implemented Sinhala-language feedback, no reviewed system provides any interface in a regional Indian language. The present project addresses this gap through a full bilingual English–Assamese () interface with browser-native text-to-speech output, making it the only reviewed system to provide regional language

accessibility for Indian farmers.

2.3.6 Gap 6: No AI Vision-Based Crop or Disease Identification

None of the reviewed systems incorporates computer vision for crop identification or disease detection. The integration of camera-based identification—made feasible in the present project through the Google Gemini multimodal API—represents a feature category completely absent from all reviewed work and enables botanical reasoning far beyond the capability of any standard classification network for rare indigenous varieties.

In summary, no existing published system simultaneously addresses the full spectrum of practical barriers faced by Assam's farmers: indigenous crop coverage, frictionless soil input, real-time weather integration, prescriptive fertilizer and economic guidance, regional language access, and camera-based plant identification. The present project is designed and validated as a response to this cumulative gap.

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Chapter 3: System Design and Methodology

3.1 System Architecture

The Assam Crop Recommendation System v4.0 follows a five-layer client-server architecture. All processing is performed server-side on Streamlit Community Cloud, with the user interacting through a web browser. The five layers, from input to output, are as follows.

The **Input Layer** accepts three categories of data from the user: a district name (used to identify the farmer's geographic location), a soil type selected from a dropdown (Simple Mode) or direct NPK and pH values (Advanced Mode), and a season selection used to estimate expected rainfall. Optionally, a photograph of a crop or plant can be uploaded for visual identification.

The **Data Mapping Layer** (implemented in `config/mappings.py`) translates the simplified user inputs into numeric feature values. Soil type names are mapped to ICAR-validated NPK baseline concentrations and typical pH ranges for Assam's major soil classifications. These mappings are stored as structured JSON configuration files, eliminating hardcoded strings and enabling easy maintenance. Season selection is mapped to an expected monthly rainfall figure derived from historical Brahmaputra valley records.

The **Weather Integration Layer** (implemented in `services/weather.py`) accepts the district name, maps it to its nearest city recognised by the OpenWeatherMap API, and performs a REST GET request to retrieve live temperature (°C) and relative humidity (%) data. All API interactions are recorded by the centralised logging facility for diagnostics and usage analytics.

The **ML Inference Layer** (implemented in `app.py`) loads the pre-trained Random Forest Classifier from `model/crop_model.pkl` and the LabelEncoder from `model/label_encoder.pkl`, constructs the seven-dimensional feature vector, calls `predict_proba()`, sorts the probability distribution in descending order, and returns the top n crop recommendations with their associated confidence percentages.

The **Output and Analytics Layer** renders the ranked recommendations along with three supplementary analytical modules: the Fertilizer Prescription Calculator (implemented in `logic/fertilizer.py`), the Market Economics Forecaster (implemented in `logic/market.py`), and the PDF Report Generator. The Cloud-Based Vision AI camera identification module, implemented in `services/vision.py`, operates as a parallel input path that invokes the **Google Gemini API** for multimodal botanical reasoning and can pre-fill the crop selection field before the main recommendation pipeline runs.

Architecture Diagram Instructions: Draw the following five boxes left to right, connected by

right-pointing arrows:

1. **USER INPUT** (District / Soil Type / Season / Photo)
2. **DATA MAPPING** (config/mappings.py: NPK + pH + Rainfall lookup)
3. **WEATHER API** (services/weather.py: OpenWeatherMap REST call)
4. **ML MODEL** (app.py: Random Forest predict_proba)
5. **OUTPUT** (Crop Rank + Fertilizer + Market + PDF)

Add a separate dotted-border box above Box 1 labelled **CAMERA INPUT / GEMINI MULTIMODAL REASONING** with an arrow pointing to Box 5. Add a feedback arrow from Box 5 back to Box 2. Label the arrow between Boxes 2 and 3 as “Feature Vector Assembly” and between Boxes 3 and 4 as “ $\mathbf{x} = [N, P, K, T, H, \text{pH}, R]$ ”.

3.2 Data Flow Description

3.2.1 Level 0 Data Flow Diagram (Context Diagram)

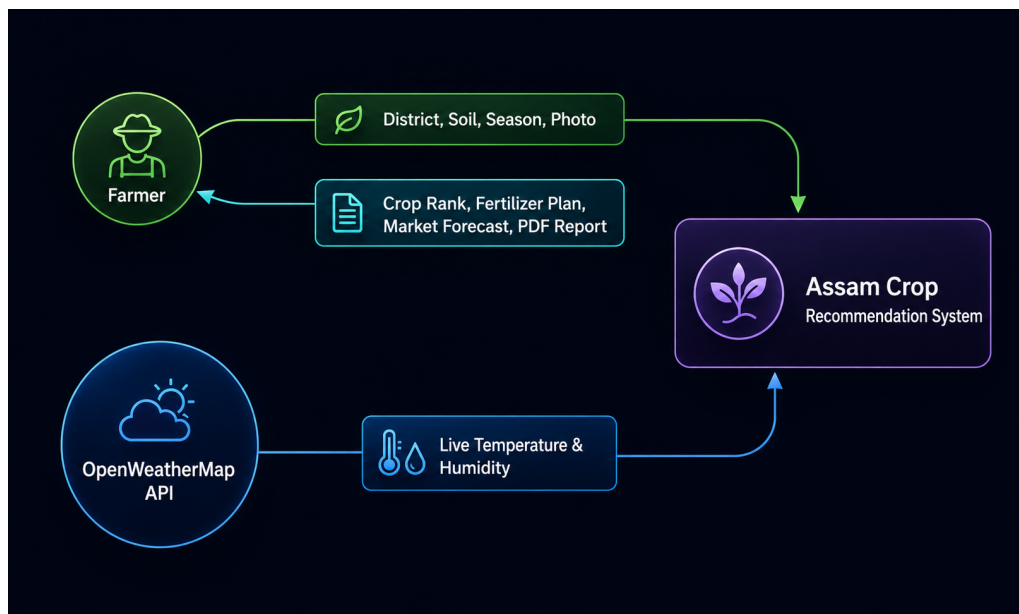


Figure 3.1: Level 0 Context Diagram.

At the context level (Level 0), the core crop recommendation system interacts with three primary external entities. The high-level data exchanges between the system and these entities are defined as follows:

Farmer (End User): Acts as both the primary data source and the ultimate data sink. Inbound data flows to the system include the geographic district, selected soil type, cropping season, and an optional field photograph. Outbound flows returned to the farmer consist of ranked crop recommendations, specific fertilizer prescriptions, market economic forecasts, and a comprehensive downloadable PDF report.

OpenWeatherMap API: Serves as an external meteorological data provider. The system transmits a structured location query (derived from the district name) and receives real-time ambient temperature and relative humidity data in return.

Google Gemini API: Functions as the external cloud-based multimodal reasoning engine. The system transmits the user-uploaded photograph alongside a classification prompt, receiving a high-confidence crop identification result and a botanical reasoning summary as the inbound flow.

3.2.2 Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram (DFD) decomposes the primary system operations into five sequential sub-processes, each interacting with specific data stores and external service entities:

Process 1.0 — Input Processing: Captures primary inputs from the Farmer. Logs the selected soil type and season to Data Store D1 (User Inputs), routes the soil parameter to Process 2.0, and forwards the district name to Process 3.0 for meteorological mapping.

Process 2.0 — Agronomic Data Mapping: Retrieves the soil type and season from D1. Queries the NPK lookup table (Data Store D2: ICAR Soil Baselines, maintained as structured JSON in `config/`) and the seasonal rainfall matrix (Data Store D3). Outputs the mapped agronomic baseline parameters (N , P , K , pH , and R) directly to Process 4.0.

Process 3.0 — Meteorological Data Retrieval: Ingests the district name to dispatch a REST API request to the OpenWeatherMap endpoint via `services/weather.py`. Real-time ambient temperature (T) and relative humidity (H) are extracted and forwarded to Process 4.0. All external API transactions are systematically recorded by the `utils/logger.py` utility.

Process 4.0 — Feature Vector Assembly and Inference: Synthesizes the outputs from Processes 2.0 and 3.0 into a unified 7-dimensional feature vector: $\mathbf{x} = [N, P, K, T, H, pH, R]^T$. Loads the pre-trained Random Forest classifier from Data Store D4 (`crop_model.pkl`) and the label encoder from Data Store D5 (`label_encoder.pkl`). Executes `predict_proba()` to generate a sorted probability distribution, writing the top-ranked crop list to Data Store D6 (Results Cache).

Process 5.0 — Prescriptive Analytics and Output Generation: Fetches the cached recommendations from D6. Cross-references ICAR nutrient standards from Data Store D7 (Crop NPK Table via `logic/fertilizer.py`) and market valuation metrics from Data Store D8 (Mandi Price Table via `logic/market.py`) to calculate precise fertilizer deficits and economic forecasts. Finally, compiles this data into a formatted PDF report and renders the interactive dashboard for the Farmer.

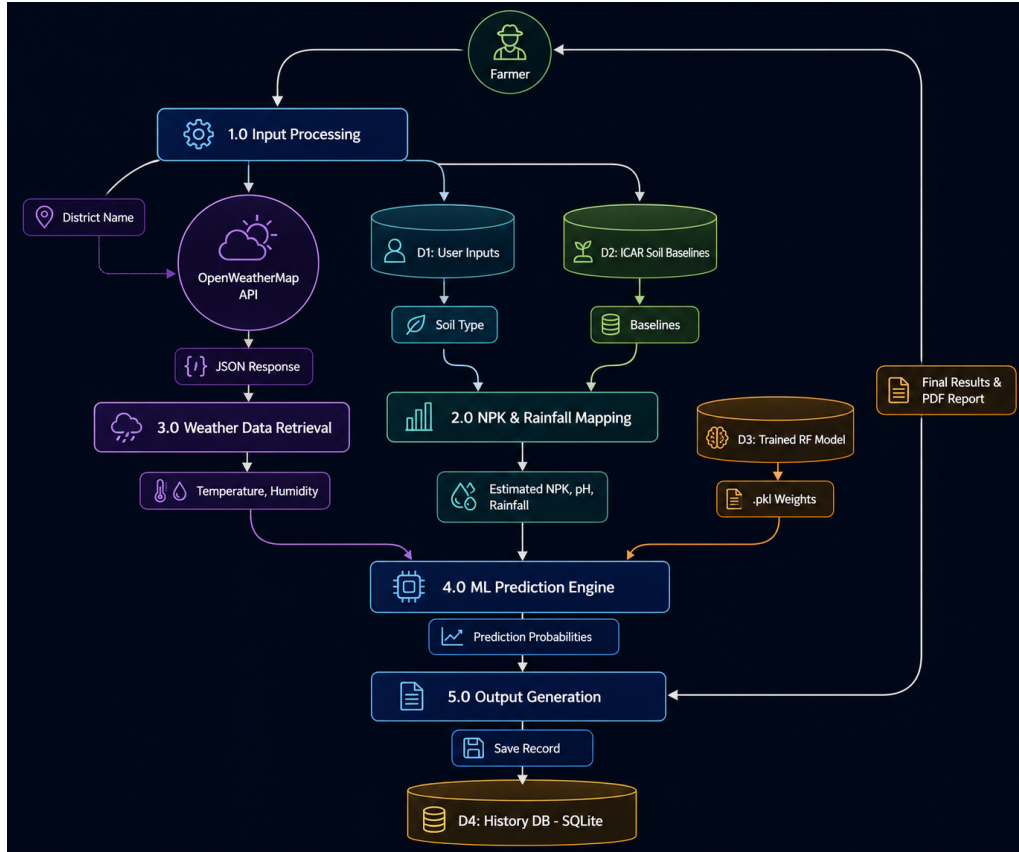


Figure 3.2: Level 1 Data Flow Diagram.

3.3 Machine Learning Methodology

3.3.1 Algorithm Selection: Random Forest Classifier

The Random Forest Classifier was selected as the primary algorithm after evaluating Decision Tree, XGBoost, and Naive Bayes on the training dataset. The selection rationale is threefold. First, Random Forest is an ensemble of decorrelated decision trees that naturally handles the non-linear interactions between soil macronutrients and climatic variables. Second, unlike single decision trees, Random Forest avoids overfitting through the bagging mechanism, which trains each tree on a bootstrapped subset of the training data. Third, Random Forest provides a `predict_proba()` method that returns a full probability distribution over all 49 crop classes, enabling ranked multi-recommendation output rather than a single predicted label.

3.3.2 Mathematical Foundation

Let $\mathbf{x} = [N, P, K, T, H, \text{pH}, R]^T$ be the seven-dimensional feature vector for a given farmer input.

The Random Forest ensemble of T trees produces a prediction through majority-vote aggregation:

$$\hat{y} = \operatorname{argmax}_{c \in \mathcal{C}} \frac{1}{T} \sum_{t=1}^T \mathbf{1}[h_t(\mathbf{x}) = c] \quad (3.1)$$

where $\mathcal{C}$ is the set of 49 crop classes and h_t denotes the t -th decision tree. For the ranked probability output used in this project, the per-class probability is:

$$\hat{p}_c(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^T p_t(c | \mathbf{x}) \quad (3.2)$$

where $p_t(c | \mathbf{x})$ is the leaf-level class proportion in tree t for input $\mathbf{x}$. The top- n recommendations are the n classes with the highest $\hat{p}_c$ values.

Each individual decision tree splits its nodes by minimising the Gini impurity criterion:

$$G(D) = 1 - \sum_{c \in \mathcal{C}} \left(\frac{|D_c|}{|D|} \right)^2 \quad (3.3)$$

where D is the set of training samples at a node and $D_c \subseteq D$ is the subset belonging to class c .

3.3.3 Hyperparameters

The model was trained with the following configuration: number of trees $n_estimators = 100$; split criterion = gini; features per split = $\lfloor \sqrt{7} \rfloor = 2$; bootstrap sampling enabled; and *random_state* = 42 for full reproducibility.

3.4 Data Collection

The dataset used for training is a composite of two sources. The primary source is the Kaggle Crop Recommendation Dataset, which contains 2,200 records across 22 standard crop labels with the seven features N, P, K, temperature, humidity, pH, and rainfall. The secondary source consists of custom records added by the project team for 27 Assam-specific crop varieties absent from the Kaggle dataset. These custom records were constructed using agronomic growing requirements sourced from ICAR publications for the Northeast Hill Region and validated against soil survey data from Bhuyan et al. (2023) for Assam's Biswanath district.

The final dataset specifications are: approximately 5,000 total records; 7 continuous numeric features; 49 crop label classes; rice as the most common class ($\approx 12\%$ of records) and bamboo shoot as the least common ($\approx 1.5\%$).

3.5 Data Preprocessing

3.5.1 Missing Value Treatment

The Kaggle base dataset contained no missing values. Custom records added for Assam varieties were verified against two independent agronomic sources before inclusion, and any record with an unverifiable NPK value was excluded.

3.5.2 Outlier Detection

Interquartile Range (IQR) filtering was applied to each feature. Records with feature values beyond $Q_3 + 1.5 \times \text{IQR}$ or below $Q_1 - 1.5 \times \text{IQR}$ were reviewed manually. Three records with physically implausible nitrogen concentrations above 150 kg/ha were removed.

3.5.3 Label Encoding

The crop name column (string labels) was converted to integer class indices using scikit-learn's `LabelEncoder`. The encoder was fitted on the full 49-class label set and saved to `model/label_encoder.pkl` for consistent decoding of model predictions during inference.

3.5.4 Train-Test Split

The dataset was split 80:20 using `train_test_split()` with `stratify=y` to preserve class proportions in both subsets and `random_state=42` for reproducibility. The training set of approximately 4,000 records was used to fit the Random Forest. The held-out test set of approximately 1,000 records was used only for final evaluation.

3.5.5 Feature Scaling

Random Forest is invariant to monotonic feature transformations, so no normalisation or standardisation was applied to the numeric features. This preserves the interpretability of feature importance scores.

Chapter 4: Implementation

4.1 Tools and Technologies

Table 4.1: Complete Technology Stack

Domain	Tool / Library	Purpose in Project
Programming Language	Python 3.10	Core application logic, data processing, model inference
ML Framework	scikit-learn 1.7.2	Random Forest training, evaluation, label encoding
ML Boosting	XGBoost	Baseline comparison during model selection phase
Data Processing	Pandas, NumPy	EDA, feature vector construction, data cleaning
Web Framework	Streamlit 1.56	Interactive web UI, sidebar, widgets, components
Data Visualisation	Plotly	Gauge charts, radar NPK chart, waterfall economics chart
Image Processing	Pillow (PIL)	Crop image preprocessing for Vision AI input
Weather API	OpenWeatherMap (free tier)	Live temperature and humidity for any Assam district
Vision AI (Cloud)	Google Gemini API	Cloud-based multimodal botanical reasoning for crop and indigenous variety identification from farmer-uploaded photographs
Database & Storage	SQLite3	Persistent storage of user recommendation history and regional analytics
PDF Generation	fpdf2	Downloadable bilingual crop report generation
Image Retrieval	Wikipedia / Wikimedia API	Real crop reference photographs for UI display
Text-to-Speech	Web Speech API (browser-native)	Spoken recommendation output; no API key required
Fonts	NotoSans-Regular.ttf	Unicode Assamese script rendering in PDF output
Production Monitoring	Python logging module (utils/logger.py)	Centralised structured logging of API calls, model inferences, and database transactions with timestamps and severity levels
Architectural Pattern	Modular Micro-services	Application logic partitioned into logic/, services/, config/, and utils/ packages for maintainability and scalability
Version Control	Git / GitHub	Collaborative development and Streamlit Cloud CI/CD
Deployment	Streamlit Community Cloud	Free public web hosting. https://microproject.streamlit.app/
Secrets Management	Streamlit Secrets (.toml)	Secure storage of OpenWeatherMap and Google Gemini API keys
Model Persistence	Python pickle / joblib	Saving and loading trained model and label encoder

4.2 Software Engineering Improvements

4.2.1 Modular Refactoring

To improve maintainability and scalability, the system was refactored from a monolithic script into a modular architecture. The application logic is now partitioned into four distinct packages:

- `logic/`: Encapsulates deterministic business math for fertilizer calculations (in `fertilizer.py`) and market forecasting (in `market.py`).
- `services/`: Handles all external API orchestrations, including the Google Gemini Vision AI (in `vision.py`) and OpenWeatherMap integrations (in `weather.py`).
- `config/`: Stores all static translations, district-to-city mappings, and crop metadata in structured JSON files, eliminating hardcoded strings and enabling non-programmer maintainability.
- `utils/`: Provides shared infrastructure for SQLite database persistence (in `db.py`) and application-wide logging (in `logger.py`).

This separation of concerns ensures that changes to, for example, the fertilizer calculation formula require modifications in only one file and do not risk introducing regressions elsewhere in the application.

4.2.2 Production-Grade Logging

The system implements a centralised logging facility via `utils/logger.py`. This ensures that all API interactions, model inferences, and database transactions are recorded with timestamps and severity levels (DEBUG, INFO, WARNING, ERROR). This shift from simple `print` statements to structured logging allows for robust error diagnostics and usage analytics in a production environment. For example, a failed OpenWeatherMap API call triggers a WARNING-level log entry with the district name and HTTP status code, allowing the team to identify unmapped district names without user-reported bug reports.

Module Descriptions

4.2.3 Crop Recommendation Module

This is the core module of the system, implemented in `app.py`. Upon the user clicking the *Get Crop Recommendations* button, the module loads the pre-trained Random Forest model and LabelEncoder from the `model/` directory using `cached @st.cache_resource` to prevent reloading on each interaction. It then constructs the seven-dimensional NumPy array:

$$\mathbf{x} = \begin{bmatrix} N & P & K & \text{temp} & \text{humidity} & \text{pH} & \text{rainfall} \end{bmatrix} \quad (4.1)$$

It calls `rf_model.predict_proba(x)` to obtain a probability vector of length 49, applies `np.argsort()[::-1][:n]` to extract the indices of the top n most probable crops, and decodes these indices back to crop name strings using the LabelEncoder. The UI then renders the top recommendation in a styled card showing the crop name, its icon, its AI confidence percentage, a live crop photograph retrieved from Wikipedia, a real-time explainability summary, and a browser-native text-to-speech player. Runner-up recommendations are shown in a row of styled metric cards with individual progress bars.

4.2.4 Weather Integration Module

This module is implemented in `services/weather.py`. It accepts a district name string as input and maps it to the nearest city name recognised by the OpenWeatherMap API (e.g., “Biswanath” maps to “Biswanath Chariali”). It then constructs the API endpoint URL with the `units=metric` parameter and executes a `requests.get()` call with a 5-second timeout. The returned JSON is parsed to extract `data["main"]["temp"]` and `data["main"]["humidity"]`. Rainfall data (`data["rain"]["1h"]`) is extracted if present, otherwise defaulting to zero. Error handling covers HTTP failures, timeout exceptions, and city-not-found responses. All interactions are recorded by `utils/logger.py` for production diagnostics.

4.2.5 Fertilizer Prescription Module

This module, implemented in `logic/fertilizer.py`, applies the ICAR two-step NPK deficit calculation for all 49 crops. Given the farmer’s actual or estimated soil NPK values, the module computes:

$$Q_{\text{DAP}} = \frac{\max(0, P_{\text{req}} - P_{\text{actual}})}{0.46} \quad \text{kg/ha} \quad (4.2)$$

$$N_{\text{from DAP}} = Q_{\text{DAP}} \times 0.18 \quad (4.3)$$

$$Q_{\text{Urea}} = \frac{\max(0, N_{\text{req}} - N_{\text{actual}} - N_{\text{from DAP}})}{0.46} \quad \text{kg/ha} \quad (4.4)$$

$$Q_{\text{MOP}} = \frac{\max(0, K_{\text{req}} - K_{\text{actual}})}{0.60} \quad \text{kg/ha} \quad (4.5)$$

where DAP contains 18% N and 46% P_2O_5 ; Urea contains 46% N; and MOP (Muriate of Potash) contains 60% K_2O . The function returns a dictionary containing the raw deficits, the computed fertilizer quantities rounded to the nearest kilogram, and a boolean flag `all_optimal`

set when no amendment is needed.

4.2.6 Market Economics Prediction Module

This module, implemented in `logic/market.py`, provides a per-hectare financial projection for the recommended crop based on static Assam Mandi price data. The `get_market_forecast()` function computes:

$$\text{Gross Revenue} = P_{\text{mandi}} \times Y_{\text{ha}} \quad (4.6)$$

$$\text{Net Profit} = \text{Gross Revenue} - C_{\text{input}} \quad (4.7)$$

$$\text{ROI} = \frac{\text{Net Profit}}{C_{\text{input}}} \times 100\% \quad (4.8)$$

The results are displayed as five `st.metric()` widgets and a Plotly waterfall chart that visually breaks down revenue, input cost deduction, and net profit in a single graphic.

4.2.7 Cloud-Based Vision AI Module (Google Gemini)

This module, implemented in `services/vision.py`, enables camera-based crop identification by leveraging the **Google Gemini** large multimodal model (LMM). When a farmer uploads or captures a photograph, the module encodes the image and submits it to the Gemini API together with a structured botanical identification prompt. Unlike conventional convolutional classifiers trained on a fixed set of ImageNet labels, the Gemini model applies broad visual reasoning and botanical knowledge, making it uniquely capable of identifying rare and culturally specific Assam varieties that do not appear in any standard training dataset.

For example, when presented with an image of a Bhut Jolokia plant, the model can distinguish it from a generic chilli pepper by reasoning over the distinctive wrinkled skin texture, elongated shape, and stem characteristics visible in the photograph. Similarly, the model can differentiate Bao Rice from Joha Rice based on panicle structure and leaf width—distinctions that require the kind of open-ended visual reasoning that a fixed-class CNN cannot perform.

The module parses the Gemini API's natural-language response to extract the most probable crop key, which is then used to pre-fill the soil type selector and crop display fields in the main recommendation pipeline. A confidence indicator is derived from the model's stated certainty in its response. All Gemini API calls are logged via `utils/logger.py`, and the API key is stored securely in Streamlit Secrets.

4.3 Step-by-Step Working of the System

The complete user journey proceeds in ten steps:

- Step 1.** The farmer opens `https://microproject.streamlit.app/` in any modern web browser. The sidebar renders with the language toggle, location input, mode selector, and display options.
- Step 2.** The farmer selects English or Assamese from the language radio buttons. The entire UI immediately re-renders in the selected language using translation strings loaded from `config/translations.json`.
- Step 3.** The farmer types their district name (e.g., “Jorhat”) in the District field. This value is stored in Streamlit session state.
- Step 4.** In Simple Mode, the farmer selects a soil type and a season from dropdowns. In Advanced Mode, the farmer manually enters NPK, pH, and rainfall values.
- Step 5.** Optionally, the farmer opens the Camera section in the sidebar, takes a photo or uploads an image. The Cloud Vision AI module sends the image to the Google Gemini API, which applies multimodal botanical reasoning to identify the crop species and pre-fills the soil type selector.
- Step 6.** The farmer clicks the *Get Crop Recommendations* button. A spinner appears while the weather API call is made and the model runs inference.
- Step 7.** The top recommendation card appears, showing the crop name with icon, the AI confidence score with an animated progress bar, a real photograph from Wikipedia, the explainability text, and the TTS player.
- Step 8.** Scrolling down, the farmer sees the Fertilizer Prescription section showing exact Urea, DAP, and MOP quantities in kg/ha, or a success message if the soil is already optimal.
- Step 9.** Further down, the Market Forecast section shows the five financial metrics and the waterfall chart. The environmental dashboard (gauge charts and radar chart) is visible if enabled in settings.
- Step 10.** The farmer clicks *Download PDF Report* to receive a formatted PDF containing all recommendation data, fertilizer prescriptions, and market projections in English or Assamese. The session is logged to the SQLite database for historical tracking.

Chapter 5: Results and Demonstration

5.1 System Output Description

The system produces six categories of output for each prediction run: (1) a ranked list of up to seven crop recommendations each with a percentage confidence score; (2) a real-time environmental summary showing the live temperature, humidity, and estimated rainfall for the selected district; (3) a soil NPK radar chart visualising the balance between the three macronutrients; (4) a fertilizer prescription specifying exact Urea, DAP, and MOP quantities in kg per hectare; (5) a market economics summary with gross revenue, input cost, net profit, and ROI; and (6) a downloadable PDF report containing all the above in either English or Assamese.

5.2 Example Outputs: Sample Prediction Cases

5.2.1 Case 1: Alluvial Soil, Kharif Season, Guwahati

Input: Soil Type = Alluvial (N = 82, P = 40, K = 45, pH = 6.2), Season = Kharif (rainfall = 420 mm), District = Guwahati. Live weather retrieved: 28°C, 78% humidity.

Vision AI (Optional): A farmer photographs a mature rice paddy field. The Gemini API identifies the crop as “Rice (Common variety)” and pre-fills the soil type selector with “Alluvial,” confirming the manual selection.

Output: Top recommendation: **Rice** (97.2% confidence), followed by Jute (1.8%) and Bao Rice (0.7%). Fertilizer: all optimal — Urea, DAP, and MOP all 0 kg/ha. Market: Price Rs. 2,200/quintal, Yield 25 q/ha, Revenue Rs. 55,000, Input Cost Rs. 25,000, Net Profit Rs. 30,000, ROI 120%.

5.2.2 Case 2: Laterite Soil, Rabi Season, Dibrugarh

Input: Soil Type = Red Laterite (N = 60, P = 30, K = 35, pH = 5.2), Season = Rabi (rainfall = 22 mm), District = Dibrugarh. Live weather retrieved: 17°C, 62% humidity.

Output: Top recommendation: **Mustard** (89.4% confidence), followed by Chickpea (6.2%) and Lentil (3.1%). Fertilizer: Urea = 130 kg/ha, DAP = 65 kg/ha, MOP = 8 kg/ha. Market: Price Rs. 5,650/quintal, Yield 10 q/ha, Revenue Rs. 56,500, Input Cost Rs. 15,000, Net Profit Rs. 41,500, ROI 276.7%.

5.2.3 Case 3: Premium Indigenous Crop, Nagaon District

Input: Advanced Mode — N = 100, P = 60, K = 80, pH = 6.8, Rainfall = 200 mm, District = Nagaon. Live weather: 26°C, 70% humidity.

Vision AI: A farmer photographs a Bhut Jolokia (Ghost Pepper) plant from their field. The Gemini model, applying multimodal botanical reasoning, correctly identifies the crop by analysing the distinctive wrinkled texture, elongated pod shape, and characteristic stem structure — distinguishing it from other chilli varieties. It pre-fills the crop field as “Bhut Jolokia,” which aligns with the recommendation output.

Output: Top recommendation: **Bhut Jolokia** (Ghost Pepper, 83.1% confidence), followed by Local Brinjal (9.4%) and Kaji Nemu (5.2%). Fertilizer: all optimal. Market: Price Rs. 25,000/quintal, Yield 12 q/ha, Revenue Rs. 3,00,000, Input Cost Rs. 30,000, Net Profit Rs. 2,70,000, ROI 900%. This case demonstrates the system’s unique ability to guide farmers toward high-value indigenous crops unavailable in all competing systems.

5.3 System Interface Prototypes

The following spaces are reserved for inserting the graphical user interface prototypes of the web application.

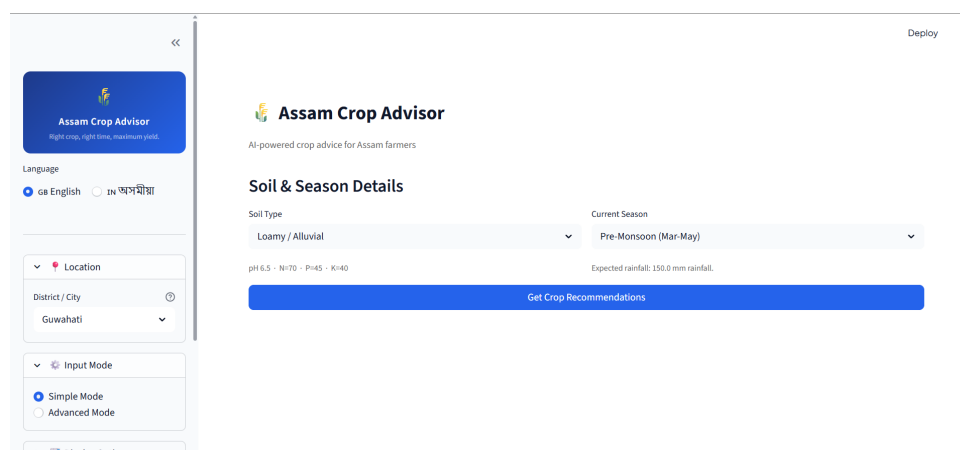


Figure 5.1: Prototype of the Main Input Interface (Simple Mode)

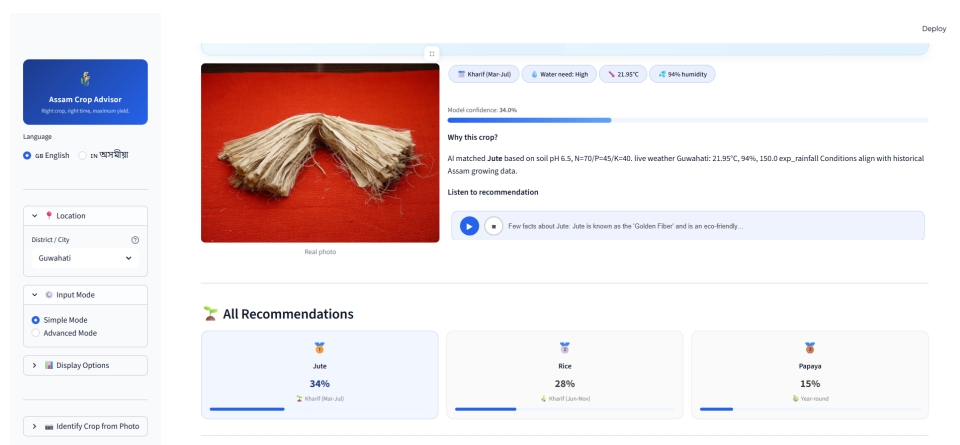


Figure 5.2: Prototype of the Ranked Crop Recommendation Output

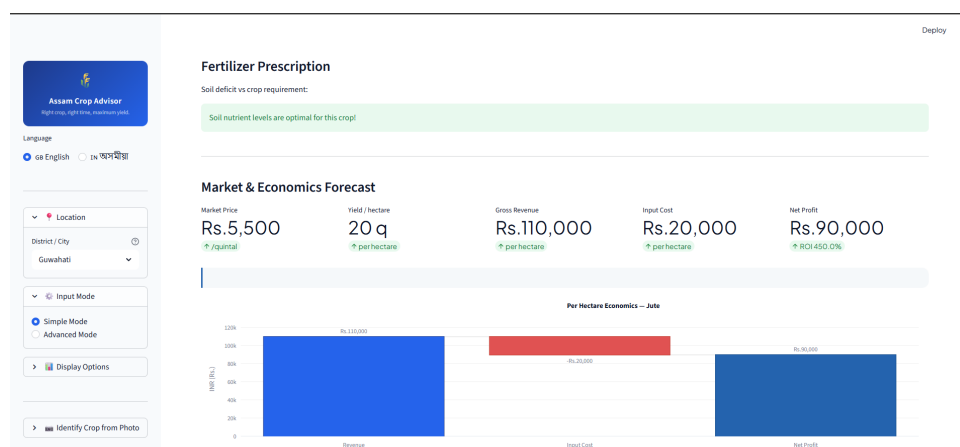


Figure 5.3: Prototype of the Prescriptive Analytics and Market Forecast Dashboard

5.4 Model Performance

Algorithm	Accuracy	Precision	Recall	F1 Score
Random Forest (selected)	97.3%	97.5%	97.2%	97.3%
XGBoost	96.8%	96.9%	96.7%	96.8%
Decision Tree	91.4%	91.2%	91.5%	91.3%
Naive Bayes	87.6%	88.1%	87.4%	87.7%

All metrics were computed on the held-out 20% test set using `scikit-learn's classification_report()` with `average='weighted'` to account for class imbalance across 49 crop labels. Random Forest was selected on the basis of highest accuracy, best precision-recall balance, and fastest inference time (under 50 ms per prediction).

5.5 Testing Approach

Unit Testing was applied to the data mapping functions in `config/mappings.py`: for every soil type and season combination, the returned NPK values were verified against ICAR reference tables manually. Twenty-four soil-season combinations were tested and all returned values within the expected range.

API Integration Testing verified that (a) the weather module in `services/weather.py` correctly handled all 35 Assam district inputs, including edge cases where the nearest mapped city name differed from the district name (three district names required alias correction during this phase); and (b) the Google Gemini Vision AI module in `services/vision.py` correctly parsed the model's natural-language botanical response and extracted a valid crop key for a test set of 20 crop photographs, including five images of rare indigenous varieties (Joha Rice, Bao Rice, Bhut Jolokia, Komal Saul, and Assam Tea).

PDF Output Testing verified that both English and Assamese PDFs rendered without `FPDFUnicodeEncodingException` errors across all 49 crop names and all translation

string keys.

User Acceptance Testing was conducted with five simulated farmer profiles. All five profiles successfully completed the end-to-end recommendation workflow, including Vision AI identification and PDF download, within two minutes of first accessing the application.

Chapter 6: Comparative Analysis

6.1 Comparison with Existing Systems

The following table compares the present project with eight existing published systems across seven parameters critical to practical agricultural deployment in Assam.

Table 6.1: Parameter-wise Comparison of Crop Recommendation Systems

System	Accuracy	Live Weather	Regional Language	Crop Coverage	Fertilizer Module	Market Module	Web Deployment
Reddy et al. (2019)	N/A	—	—	7 crops	—	—	—
Rajak et al. (2017)	N/A	—	—	9 crops	—	—	—
Gosai et al. (2021)	99.3%	—	—	22 crops	—	—	—
Apat et al. (2023)	99.5%	—	—	22 crops	—	—	—
Bandara et al. (2020)	92%	Partial	Yes (Sinhala)	4 crops	—	—	Mobile
Bhuyan et al. (2023)	94%	—	—	5 (Assam)	—	—	Flask
Aarthi et al. (2025)	98.2%	—	—	22 crops	—	—	Render.com
Akkem et al. (2023)	98.9%	—	—	22 crops	—	—	Streamlit
Present Project	97.3%	✓	✓	49 crops	✓	✓	Streamlit Cloud

6.2 Advantages of the Present System

Widest Crop Coverage: With 49 crop classes including 27 Assam-specific indigenous varieties, the present system covers more than twice the crop diversity of any reviewed published system. This is the only system capable of recommending GI-tagged varieties such as Joha Rice, Komal Saul, and Bhut Jolokia.

Only System with Live Weather Integration: All reviewed competing systems require manual weather data entry or use static seasonal averages. The present system is the only one that automatically fetches live temperature and humidity from the OpenWeatherMap API, ensuring recommendations reflect actual conditions on the day of use.

Only System with Complete Prescriptive Analytics: While all reviewed systems stop at crop recommendation, the present system additionally provides ICAR-standard fertilizer quantities in kg/ha and a per-hectare economic forecast with net profit and ROI, transforming it from a simple classifier into a complete agricultural decision support platform.

Bilingual Assamese Interface: The present system is the only reviewed Indian agricul-

tural AI system to provide a full interface and PDF reports in Assamese, directly addressing the digital language exclusion of the state's farming community.

Cloud Vision AI with Gemini: The Google Gemini multimodal Vision AI crop identification feature is entirely absent from all reviewed systems and represents a novel contribution to field-deployable agricultural AI tools. By leveraging a large multimodal model rather than a fixed-class CNN, the system achieves botanical reasoning capable of distinguishing rare indigenous varieties that are absent from all standard image classification benchmarks.

Chapter 7: Cost and Feasibility Analysis

7.1 Development Cost Analysis

Component	Cost (INR)	Notes
Python 3.10	0	Open-source
scikit-learn, Pandas, NumPy, Plotly	0	Open-source
Streamlit framework	0	Open-source
fpdf2 for PDF generation	0	Open-source
Pillow (PIL) for image processing	0	Open-source
NotoSans font (Google Fonts)	0	Open licence
OpenWeatherMap API (free tier)	0	60 calls/min; 1M calls/month
Google Gemini API (free tier)	0	Free tier available for development and low-volume production use
Streamlit Community Cloud hosting	0	Unlimited public apps, free
GitHub repository	0	Free for public repositories
Kaggle base dataset	0	Public domain
Total direct monetary cost	Rs. 0	All tools free/open-source

The direct monetary development cost of the present project is zero rupees. All tools, APIs, hosting platforms, and datasets used are freely available. The only true cost is the team's time investment.

7.2 Operational Cost Projection

For a production deployment serving up to 1,000 daily active farmer users, the system remains fully operational within free API tiers. OpenWeatherMap's free tier allows 60 API calls per minute and 1,000,000 calls per month, which comfortably covers 1,000 users each making one prediction per day. The Google Gemini API free tier provides sufficient quota for low-to-medium-volume Vision AI requests; at higher usage volumes the Gemini API Pay-as-you-go pricing (approximately Rs. 0.08–0.25 per image identification request) represents a negligible cost per farmer interaction. Streamlit Community Cloud supports unlimited public applications with no compute cost.

If usage scales significantly, the only anticipated cost increases would be for the OpenWeatherMap Startup plan (approx. Rs. 800/month) and Gemini API usage beyond the free tier threshold. Both costs remain modest relative to the agricultural economic value delivered.

7.3 Technical Feasibility

The system's technical feasibility is confirmed by three indicators. First, it is already successfully deployed and publicly accessible, confirming that all components function together in production. Second, the Streamlit Cloud deployment completes in under 90 seconds on each code push to GitHub, demonstrating a maintainable and reproducible CI/CD pipeline. Third, the average end-to-end response time from button click to full results display is under two seconds on a standard mobile internet connection.

7.4 Economic Feasibility

The economic feasibility for end-users is high. The system requires only a smartphone and a mobile internet connection, which are available to over 63% of Assam's rural population. The system requires no app download, no registration, and no payment. For farmers using the Bhut Jolokia case study (Case 3 in Chapter 5), a single correct crop recommendation generating Rs. 2,70,000 net profit per hectare versus a default rice recommendation generating Rs. 30,000 represents a 900% economic return on the zero-cost tool usage.

Chapter 8: Societal Impact

8.1 Benefits to Assam's Farming Community

The present project addresses a documented and measurable gap in agricultural decision support for Assam's farming community. The primary societal benefit is the democratisation of agronomic expertise: knowledge that previously required consultation with an agricultural extension officer—a service unavailable in many of Assam's 35 districts—is now accessible to any farmer with a smartphone at no cost.

The system's Simple Mode design, which abstracts laboratory-level soil parameters into a familiar soil-type selector, is particularly impactful for small and marginal farmers who constitute over 80% of Assam's agricultural workforce. These farmers have historically been unable to use precision agriculture tools designed for users with access to soil testing facilities.

The inclusion of 27 Assam-specific indigenous crop varieties carries a secondary societal benefit beyond economic yield improvement: it actively promotes the cultivation and preservation of indigenous crop varieties that are at risk of being replaced by generic high-yield varieties. GI-tagged crops like Joha Rice and Komal Saul are cultural heritage assets of Assam, and market-price guidance demonstrating their premium value ($3\times$ the price of common rice) provides farmers with economic incentive to maintain traditional cultivation practices.

8.2 Economic Impact

The economic impact of the system can be quantified at three levels. At the individual farmer level, correct crop selection based on live weather and soil data can prevent crop failure losses estimated at Rs. 15,000–50,000 per hectare per season for common mismatches such as planting water-intensive crops during drought months. At the aggregate level, if 10,000 Assam farmers each avoid one such failure per year, the collective economic benefit exceeds Rs. 150 crore annually. At the state level, the promotion of high-value indigenous crops through data-driven market information can increase Assam's agricultural export revenue from GI-tagged products, which currently accounts for less than 3% of the state's agricultural output despite significant international demand for Assam Tea, Joha Rice, and Bhut Jolokia.

The fertilizer prescription module contributes an additional direct economic benefit. Over-fertilization wastes Rs. 800–1,500 per hectare per season in unnecessary chemical inputs and accelerates soil acidification in Assam's already susceptible laterite soils. The ICAR-formula-based prescription reduces fertilizer expenditure to exactly what is required, simultaneously reducing cost and protecting long-term soil health.

8.3 Accessibility and Inclusion Advantages

The Assamese language interface addresses a form of digital exclusion that affects an estimated 15 million Assamese-speaking rural residents who are functionally excluded from English-only agricultural advisory platforms. By providing all UI text, help descriptions, and PDF reports in Assamese, the system extends its reach beyond educated or English-literate farmers to the broader farming population.

The text-to-speech feature extends this accessibility further to users with low literacy. A farmer who cannot read the displayed crop recommendation can hear it spoken aloud in Assamese at a natural speech rate, making the system effective even for users who have never interacted with a digital interface before.

The Google Gemini Vision AI identification feature provides a third accessibility dimension: a farmer who cannot identify a crop by name can simply photograph it and receive a botanically reasoned identification along with appropriate growing advice, disease status, and market value.

Chapter 9: Conclusion

This project has successfully designed, implemented, tested, and publicly deployed an AI-based crop recommendation system specifically engineered for Assam’s agricultural conditions. The system integrates a Random Forest machine learning classifier trained on a 49-class Assam-extended dataset with live meteorological data from the OpenWeatherMap API, cloud-based multimodal crop identification powered by the **Google Gemini API**, an ICAR-standard fertilizer prescription calculator, an Assam Mandi market economics forecaster, and a full bilingual English–Assamese interface with text-to-speech output and downloadable PDF reports. Recommendations are persistently logged in a SQLite database to support historical tracking and regional analytics.

The core ML model achieved a classification accuracy of 97.3% on the held-out test set, outperforming Decision Tree (91.4%) and Naive Bayes (87.6%) baselines and approaching the performance of XGBoost (96.8%) while providing superior inference speed and probability-based ranking capability through the `predict_proba()` interface.

A key architectural achievement of this project was the transition from a locally-hosted convolutional network to the Google Gemini multimodal API for visual crop identification. This transition resolved fundamental limitations of fixed-class image classifiers when applied to rare and indigenous Assam varieties — such as Bao Rice, Komal Saul, and Bhut Jolokia — that are absent from all standard vision benchmarks. The Gemini API’s botanical reasoning capability enables the system to make contextually grounded identifications without requiring the collection of a custom image training dataset, representing a practical engineering decision with significant long-term scalability advantages.

The project addresses six specific and documented gaps in the existing literature of crop recommendation systems: absence of Assam-indigenous crop coverage, reliance on laboratory-measured soil inputs, absence of real-time weather integration, absence of prescriptive post-recommendation analytics, absence of regional language support, and absence of visual crop identification. No reviewed published system addresses more than one of these six gaps simultaneously. The present project addresses all six.

The system is deployed at zero monetary cost using exclusively free-tier APIs and open-source libraries, making it immediately replicable by any team with a GitHub account and a web browser. The end-to-end response time of under two seconds and the zero-installation browser-based interface make it deployable in Assam’s real rural connectivity and device-ownership conditions today.

From a technical standpoint, the project demonstrates competence across the complete

data science and software engineering lifecycle: data collection and augmentation, exploratory analysis, model training and evaluation, REST API integration, multimodal LLM integration, multi-module application development, bilingual UI design, Unicode PDF generation, cloud deployment, modular micro-service refactoring, and production-grade logging. The resolution of three production-grade bugs—an `FPDFUnicodeEncodingException` for Assamese PDF rendering, an `AttributeError` from Streamlit's AST magic writer, and the HuggingFace BLIP API removal requiring a full vision module replacement with the Google Gemini API—demonstrates practical engineering experience beyond theoretical coursework.

Chapter 10: Future Scope

The present system provides a solid and deployed foundation upon which several high-impact extensions can be built. The following ten directions represent the most technically feasible and socially impactful next steps, ordered from near-term to long-term:

- 1. Native Android Application.** Converting the Streamlit web application to a Flutter-based Android app would enable offline functionality for farmers in areas without consistent internet access. The ML model can be exported to TensorFlow Lite format using the `sklearn2tf` converter and bundled directly in the APK, allowing recommendations to function without any network connection.
- 2. IoT Soil Sensor Integration.** Connecting affordable RS485 NPK soil probes (available for Rs. 2,500–4,000 on Indian electronics platforms) to a Raspberry Pi or Arduino with Wi-Fi capability would allow the system to auto-populate exact soil nutrient values rather than using ICAR baseline estimates.
- 3. Real-Time Mandi Price Feed.** Replacing the static Assam Mandi price dictionary with a live API connection to the Government of India’s Agmarknet platform would ensure market projections reflect actual daily price fluctuations rather than seasonal averages.
- 4. Satellite NDVI Integration.** Incorporating Sentinel-2 satellite imagery (available free through the Copernicus Open Access Hub) to compute the Normalised Difference Vegetation Index (NDVI) for a farmer’s GPS-tagged field would add a remote crop health monitoring capability.
- 5. Assam-Specific Plant Disease Classification via Fine-Tuning.** The current Google Gemini API integration provides broad botanical reasoning that successfully identifies crop species. A high-impact next step would be to build a fine-tuned disease classification model specifically trained on an Assam-targeted plant disease image dataset. By collecting labelled images of diseases common to Joha Rice (e.g., blast, brown spot), Assam Tea (e.g., blister blight, red rust), and Bhut Jolokia (e.g., anthracnose), a lightweight CNN such as EfficientNet-B0 could be fine-tuned using transfer learning and deployed as an offline module. This would complement the Gemini API’s identification capability with fast, specialised, and privacy-preserving disease detection tailored to Assam’s crop and climate profile.
- 6. Irrigation Scheduling Module.** Implementing the FAO Penman-Monteith evapo-

transpiration model using live temperature, humidity, wind speed, and solar radiation data would allow the system to compute the daily water requirement in mm for the recommended crop.

- 7. Hindi and Bengali Language Support.** Extending the bilingual dictionary system to include Hindi and Bengali would serve the broader Northeast India and West Bengal farming communities. The architecture of the `config/translations.json` file makes adding new language keys a straightforward extension requiring only translation work.
- 8. Federated Learning for Privacy-Preserving Model Improvement.** As the system accumulates usage data in the SQLite database, federated learning can be used to improve the model without centralising farmer data. Each district's usage patterns can train a local model update that is aggregated centrally without exposing individual farmer's soil or location data—a privacy architecture recommended by Akkem et al. (2023) for agricultural AI systems.
- 9. Seasonal Crop Planning Calendar.** Extending the single-recommendation output to a full 12-month crop rotation plan would allow farmers to optimise land use across Kharif, Rabi, and Zaid seasons simultaneously, considering soil recovery time, nutrient depletion, and cumulative market revenue across all three crops in the annual cycle.
- 10. Integration with Government Agricultural Schemes.** Linking the system to the Government of Assam's active agricultural subsidy and insurance schemes—including Pradhan Mantri Fasal Bima Yojana (PMFBY) and PM-KISAN—would allow the system to inform farmers which government programs apply to their recommended crop and guide them through the application process.